COGGING TORQUE COMPARISON

FEMM versus Magneto2D

12 registered fixtures | fresh Windows measurements | proposed protocol 1.7.0 | no fitting or waveform alignment

Baseline results: 9 cases pass every proposed gate. Three distributed cases meet both amplitude limits, but their baseline FEMM references fail zero-mean and/or period-closure checks.

Refinement update: all three finer FEMM references now pass both self-checks. Magneto2D versus these finer references: 1 PASS, 2 YELLOW. See page 7.

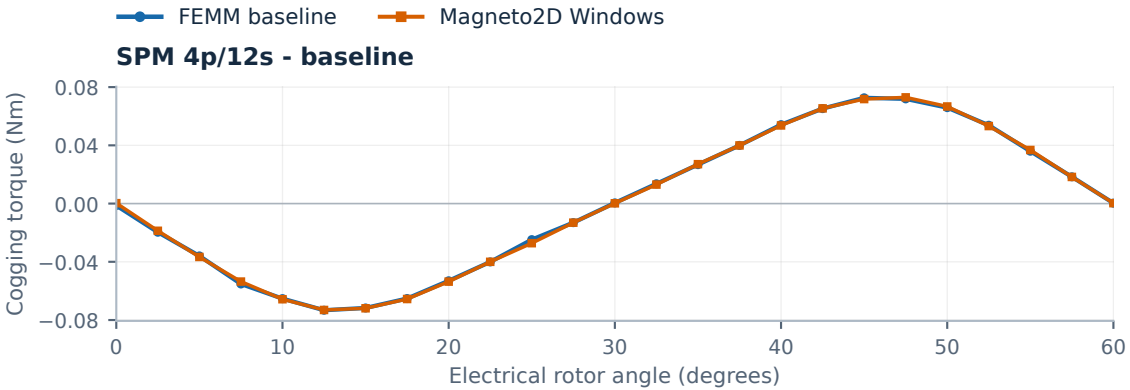
Fixture	FEMM peak	M2D peak	FEMM p-p	M2D p-p	Baseline gates
SPM 4p/12s - baseline	0.073	0.073	0.146	0.146	PASS
SPM 4p/12s - tight air gap	0.242	0.243	0.481	0.485	PASS
SPM 8p/12s - baseline	0.426	0.450	0.849	0.899	PASS
SPM 8p/12s - tight air gap	0.697	0.733	1.392	1.464	PASS
SPM 8p/12s - shoed example	0.657	0.678	1.301	1.332	PASS
IPM 8p/12s - FSAE	0.133	0.135	0.266	0.268	PASS
IPM 8p/12s - angled pocket	0.283	0.281	0.565	0.560	PASS
IPM 10p/12s - FSCW	0.522	0.539	1.016	1.076	PASS
IPM 16p/24s - industrial	0.840	0.819	1.676	1.635	PASS
SPM 8p/24s - distributed	12.192	13.188	24.035	26.344	REF REVIEW
SPM 8p/48s - distributed	8.285	8.938	16.487	17.827	REF REVIEW
SPM 12p/36s - distributed	12.368	13.557	24.568	27.010	REF REVIEW

All torque values in Nm. Peak = max |T| over the 25 sampled positions. Peak-to-peak (p-p) = max T - min T. Peaks are sampled values; extrema between samples are not inferred.

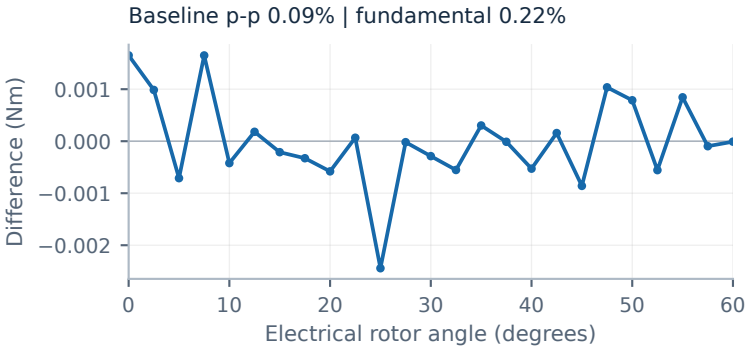
Pages 2-5 plot all fixtures. Dashed green lines show completed finer FEMM runs. Baseline comparisons and verdicts are preserved. Page 6 details the original failures; page 7 gives the refinement results.

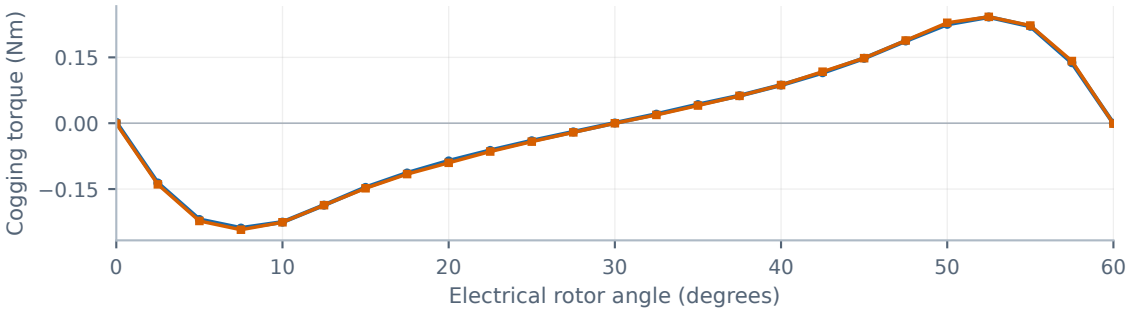
Standard SPM fixtures

Signed torque over one cogging period; residual = Magneto2D - FEMM at identical sampled angles.

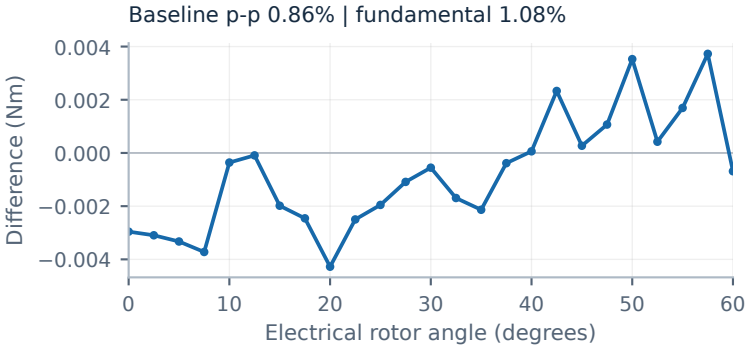


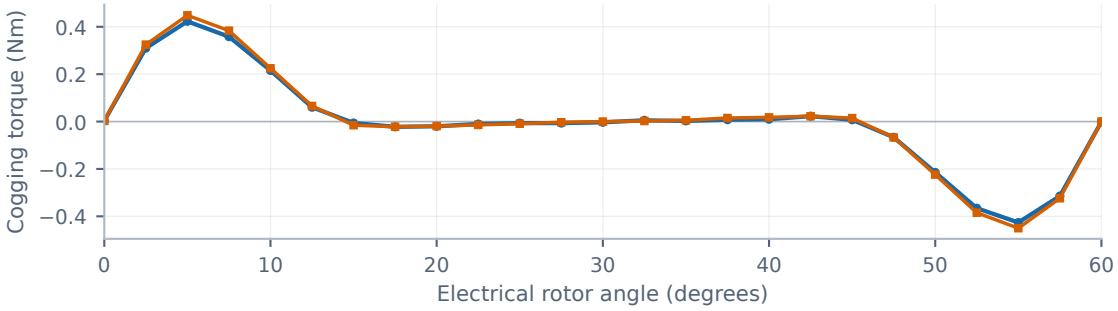
SPM 4p/12s - baseline



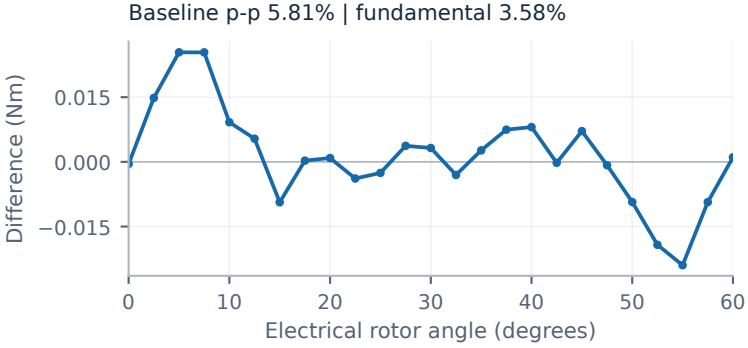


SPM 4p/12s - tight air gap





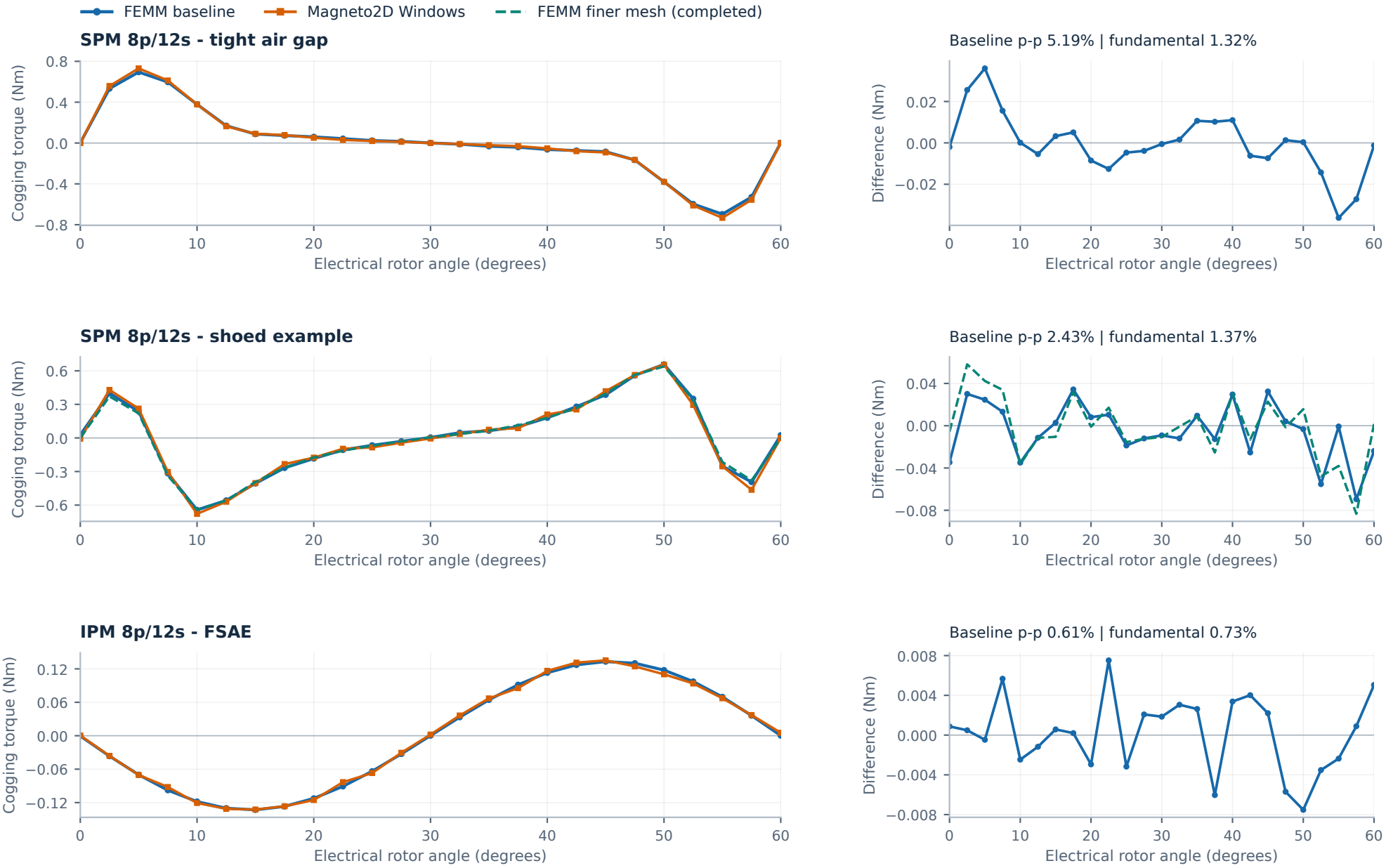
SPM 8p/12s - baseline



Baseline proposed gate status: PASS for every case on this page.

SPM and FSAE fixtures

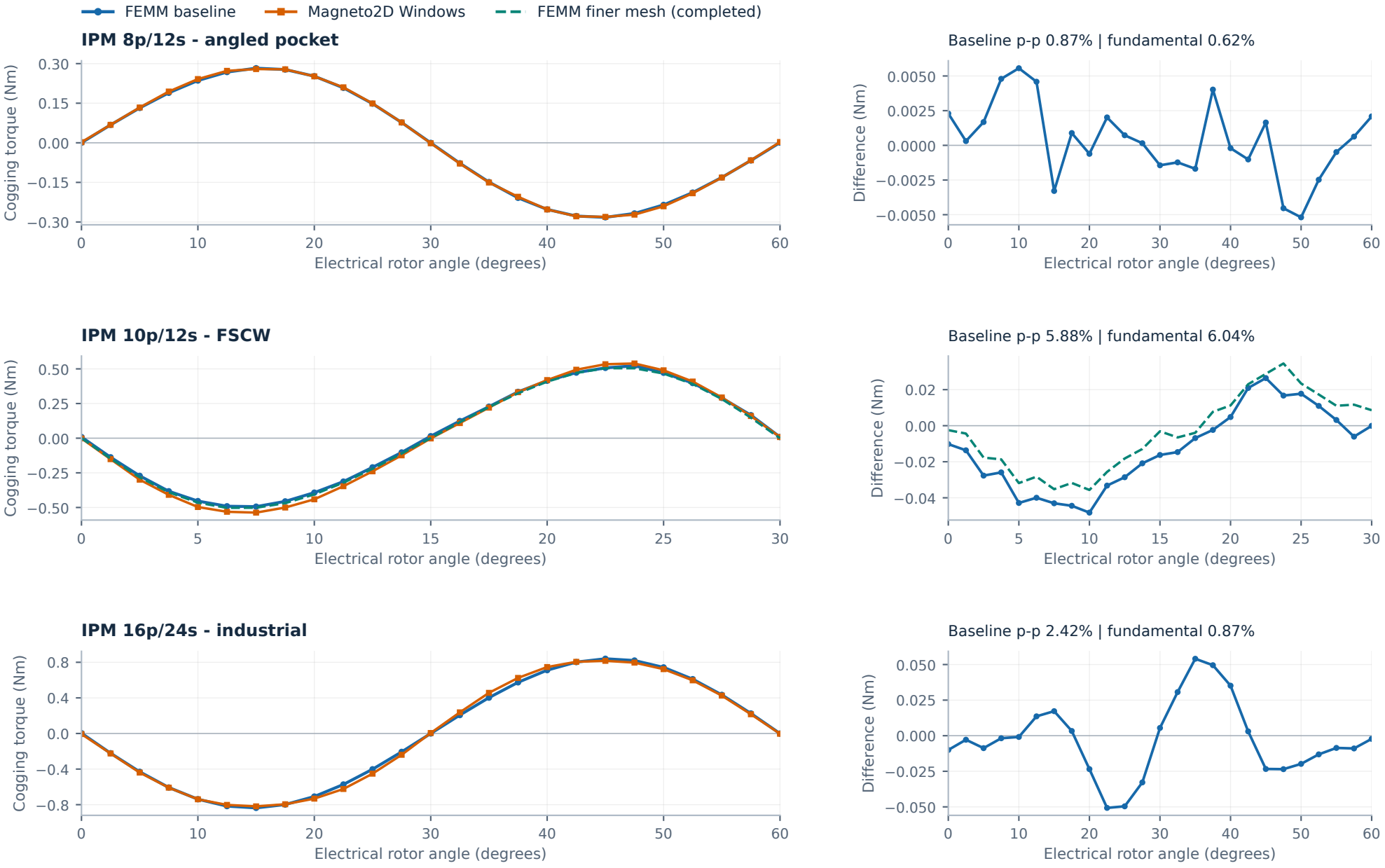
Signed torque over one cogging period; residual = Magneto2D - FEMM at identical sampled angles.



Baseline proposed gate status: PASS for every case on this page.

IPM fixtures

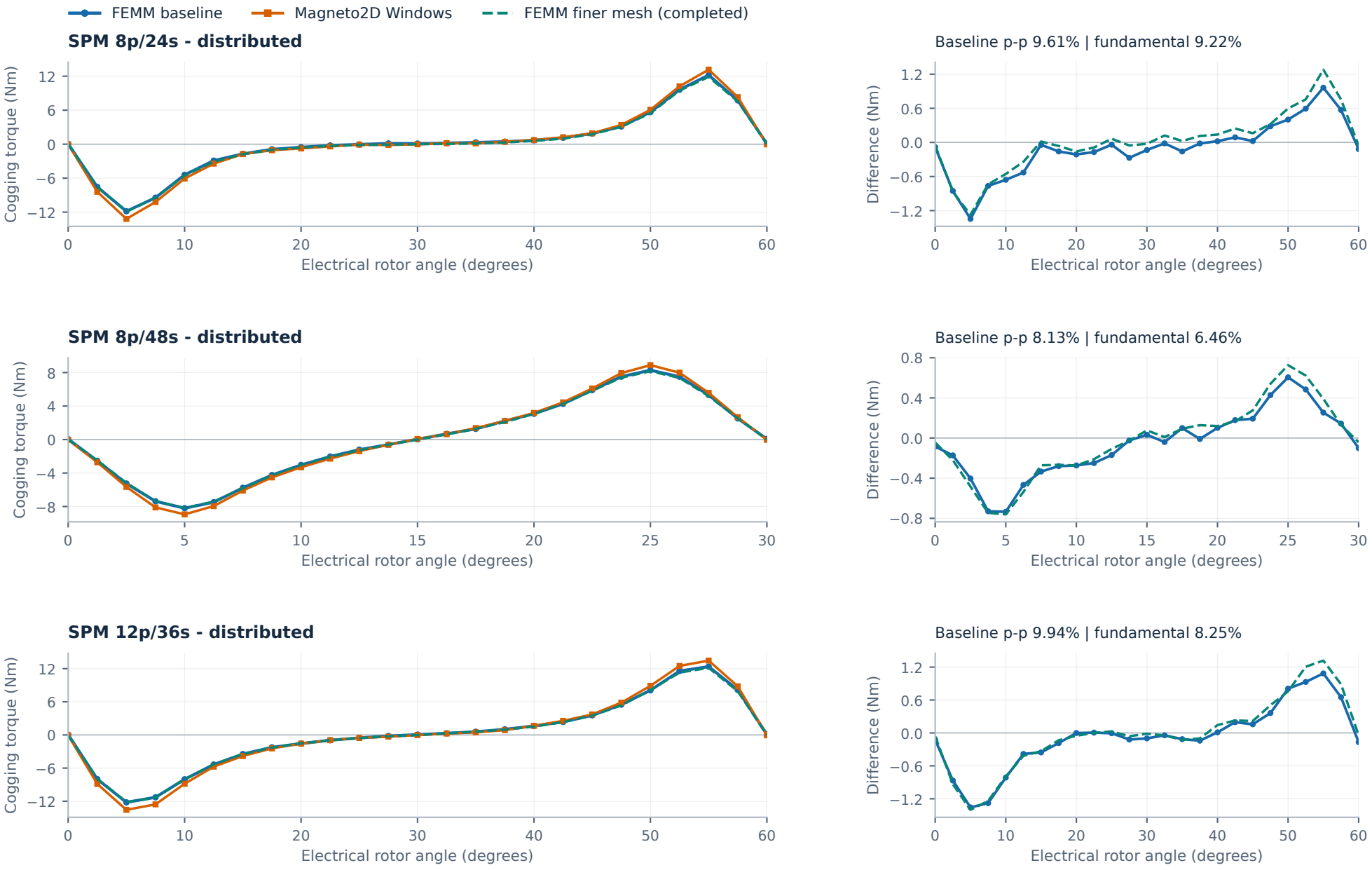
Signed torque over one cogging period; residual = Magneto2D - FEMM at identical sampled angles.



Baseline proposed gate status: PASS for every case on this page.

Distributed-winding fixtures

Signed torque over one cogging period; residual = Magneto2D - FEMM at identical sampled angles.



Finer FEMM self-checks: all PASS. Native versus finer FEMM: 1 PASS, 2 YELLOW (page 7).

Why three cases remain under review

The baseline FEMM self-checks are independent of the Magneto2D amplitude comparison.

An ideal no-current cogging waveform averages to zero over a complete period and repeats at its end. The proposed protocol bounds both residuals by 0.5% of the base fixture's FEMM mean loaded torque. All values below are Nm.

Distributed fixture	Allowed residual	Mean offset	End mismatch	Failed baseline checks
SPM 8p/24s - distributed	0.02831	0.11260	0.03345	mean + period
SPM 8p/48s - distributed	0.03152	0.03861	0.01592	mean only
SPM 12p/36s - distributed	0.02607	0.05721	0.04245	mean + period

Interpretation. The distributed cases meet the 10% peak-to-peak and fundamental-harmonic comparison limits, with peak-to-peak differences of 8.13-9.94%. Their waveform correlations exceed 0.9998. The reference residuals nevertheless exceed the registered limits. Independent remeshing or torque-extraction bias is a hypothesis; the baseline data alone does not establish the cause.

Refinement snapshot. 3/3 additional distributed finer-mesh sweeps were complete when this document was generated (03 Nov 2026, 07:01 PM EST). Incomplete sweeps are omitted. The two original mesh controls (shoed SPM and 10p/12s IPM) are complete and pass; their baseline-to-fine p-p changes are 1.60% and 0.85%.

Measurement method. 25 positions including the repeated endpoint, over one cogging period. FEMM 4.2 uses separate static models and rotor-body weighted-stress torque, zero winding current, precision 1e-8, and a baseline gap mesh target min(0.25 mm, gap/4). Finer runs halve that target. Magneto2D uses the PR's public native Windows path and registered requests. Both use the public clockwise angle and torque convention. No alignment, filtering, fitting, or rescaling was applied.

Traceability. PR 186 at 6bc935d0beeaad1390134670a9813a7e04c3b22c. FEMM 4.2 reference collection with 425 retained no-current fields. M350-50A curve and N42 magnet contract are pinned by protocol 1.7.0. Native request/result identities and hashes and FEMM input/result hashes were checked for this comparison. Public numerical data and measured input hashes accompany this report.

Status. Proposed numerical comparison only. The reference bundle remains under review; these results confer no release qualification or hardware validation.

COMPLETED DISTRIBUTED REFINEMENT

Finer meshes resolve the FEMM self-checks

75 additional fields | same physical collector and unchanged requests | half the baseline air-gap mesh target

All three finer FEMM references pass zero mean and period closure. FEMM amplitudes decrease slightly on refinement, moving two Magneto2D comparisons into the unchanged 10-20% YELLOW band.

Distributed fixture	Fine peak	Fine p-p	M2D p-p diff	M2D H1 diff	Fine comparison
SPM 8p/24s - distributed	11.916	23.792	10.73%	10.17%	YELLOW
SPM 8p/48s - distributed	8.177	16.341	9.10%	7.30%	PASS
SPM 12p/36s - distributed	12.155	24.290	11.20%	9.24%	YELLOW

Peak and p-p are Nm; H1 is the fundamental cogging harmonic. Differences are absolute relative to finer FEMM. PASS is within max(10%, floor); YELLOW is within max(20%, twice the floor).

Distributed fixture	Allowed residual	Fine mean offset	Fine end mismatch	FEMM self-checks
SPM 8p/24s - distributed	0.02831	-0.00754	0.00341	PASS / PASS
SPM 8p/48s - distributed	0.03152	0.00603	0.00350	PASS / PASS
SPM 12p/36s - distributed	0.02607	0.00439	0.01536	PASS / PASS

Mesh sensitivity. Halving the FEMM gap target changes peak-to-peak torque by 0.90-1.15% and the fundamental by 0.79-0.92%. The much smaller mean offsets and end mismatches support a mesh-dependent explanation for the original reference consistency failures.

Remaining discrepancy. Magneto2D predicts higher amplitudes for all three distributed fixtures. The source of that roughly 7-11% difference has not been isolated. Reference improvement does not by itself establish native mesh convergence. All 75 refined FEM/ANS file pairs and zero-current audits were verified. These diagnostic results do not register or qualify a reference bundle.